# More Than 67.31 Percent of the Zeta Zeros Are Simple and on the Critical Line

Deep Bhattacharjee [1,2,*] and Ushashi Bhattacharya [3,†]

[1] *Formerly, Electro-Gravitational Space Propulsion Laboratory (EGSPL), Bhubaneswar, Odisha 751030, India*
[2] *Formerly, University of Calcutta, Senate House, 87/1 College Street, Kolkata 700073, West Bengal, India*
[3] *Formerly, National Taiwan University, Taipei 106319, Taiwan*

(Dated: 19 October 2026)

*We prove that more than 67.31 percent of the nontrivial zeros of the Riemann zeta function, counted with multiplicity, are simple and lie on the critical line, and that the method cannot reach 67.36 percent. The new ingredient is a sharp bound for the largest eigenvalue of the overlap matrix of well-separated zeros, which lets long blocks of zeros be used in full.*

*2020 Mathematics Subject Classification.* Primary 11M26; Secondary 11M06, 15A42, 42A05, 65G30.

Keywords: Riemann zeta function; simple zeros; pair correlation; rank–trace inequality

## CONTENTS

I. Introduction 1
II. The framework 2
III. The stability defect 3
IV. The overlap kernel 4
V. Separated zeros 4
VI. The defect of a block of zeros 5
VII. Eight consecutive zeros 6
VIII. Proof of Theorem 1 7
IX. How far the method goes 7
Data and code availability 8
Ethical declarations 8
A. Elementary estimates 8
B. Verification of Proposition 11 9
References 9

## I. INTRODUCTION

Write $\rho = \beta + i\gamma$ for a nontrivial zero of the Riemann zeta function, let $N(T_1, T_2)$ be the number of zeros with $T_1 < \gamma \le T_2$, counted with multiplicity, and let $N_0^s(T_1, T_2)$ be the number of those zeros that are simple and lie on the critical line $\beta = \frac{1}{2}$. That a positive proportion of the zeros lie on the line is due to Selberg [19]. Levinson's method [15] gave one third, and the zeros it detects are simple [12]; Conrey [11] passed two fifths, Bui, Conrey and Young [9] passed 41 percent, and Pratt, Robles, Zaharescu and Zeindler [18] passed five twelfths.[1]

Montgomery [16] took a different route. Assuming the Riemann hypothesis, he evaluated the pair correlation of the zeros against test functions whose Fourier transform is supported in $[-1, 1]$, and deduced that at least two thirds of the zeros are simple. Montgomery and Taylor [17] optimised the test function and reached

$$H_{\mathrm{MT}} := \frac{3}{2} - \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}} = 0.6725007036\ldots, \qquad (1)$$

which is optimal among certificates of this shape [10].[2]

In 2026 Alpöge and Furman [2] removed the Riemann hypothesis from Montgomery's deduction. They compress Weil's Hermitian form to a family of modulated copies of a window, split the resulting matrix into a positive part carried by the zeros on the line and a part of controlled positive index carried by the off-line pairs, and bound the number of simple zeros on the line by a rank–trace inequality; the analytic input is the unconditional pair correlation of Baluyot, Goldston, Suriajaya and Turnage-Butterbaugh [3]. This gives $\frac{2}{3}$, and $H_{\mathrm{MT}}$ with the Montgomery–Taylor window, unconditionally.[3] A refinement of the rank–trace inequality, which retains a nonnegative

* itsdeep@live.com; ∗Corresponding author. Second address: d.bhattacharjee@erl-forschung.de.
† rai.bhattacharya.in@gmail.com

[1] All of these results come from mollified moments of $\zeta$ or of a combination of its derivatives on the line. The mollifier is a Dirichlet polynomial chosen to cancel the large values of $\zeta$, and the proportion grows with its length; the length is limited by the available estimates for sums of Kloosterman sums. Selberg's original argument already gave a positive proportion, without an explicit constant; Levinson's constant is slightly above 0.3474, and the improvements since then each came from a longer or more flexible mollifier.

[2] Montgomery's form factor $F(\alpha)$ equals $|\alpha| + o(1)$ for $|\alpha| \le 1$; the zero side of his identity is a sum over pairs of zeros weighted by a Fourier transform, and on the Riemann hypothesis every ordinate is real, so a positive test function gives a sum of positive terms. The window that attains $H_{\mathrm{MT}}$ is $\cos(\sqrt{2}t)$ on $[-\frac{1}{2}, \frac{1}{2}]$, and the constant is $2 - R(\psi)$, where $R(\psi)$ is the normalised second moment of the window; see [2, Lemma 5.6].

[3] The paper states that the argument was found and written by Claude, a language model developed by Anthropic, and that the authors verified it; it is accompanied by a Lean 4 formalisation. Lamzouri [14] gave a second proof of the bound $\frac{2}{3}$.

stability defect of the Gram matrix of the simple zeros, and a certified inequality for seven consecutive zeros, raised the constant to 0.6730085 [1]. Our result is the following.

**Theorem 1.** *As $T \to \infty$,*

$$\liminf_{T\to\infty} \frac{N_0^s(T,2T)}{N(T,2T)} \geq \frac{3125}{3112}\Big(H_{\mathrm{MT}} - \frac{7}{3200}\Big) = 0.6731133552\ldots.$$

Summing over dyadic intervals gives the same constant for $N_0^s(0,T)/N(0,T)$. Figure 1 places the bound among the earlier ones.

The gain over [1] comes from using the stability defect in full. For a positive semidefinite matrix $M$ the defect is $\mathscr{D}(M) = \operatorname{tr}\Psi(M)$, with $\Psi(t) = (t-1)^2$ on $[0,2]$ and $\Psi(t) = 2t-3$ beyond. When every eigenvalue of $M$ is at most 2, the defect equals $\operatorname{tr}(M-I)^2$, which for a Gram matrix with unit diagonal is twice the sum of the squared overlaps of the zeros. Without control of the eigenvalues, the defect of a block of zeros can only be bounded by 1, and in [1] this cap limits the size of the blocks and hence the gain. We prove that the overlap matrix of points at mutual distance at least $\frac{11}{20}$ of the mean spacing has every eigenvalue at most 1.97903 (Lemma 8). The bound is sharp: the eigenvalues of evenly spaced points reach it, and points spaced more closely than 0.5442 produce eigenvalues above 2. Zeros closer than $\frac{11}{20}$ are treated separately: they overlap so strongly that a pair of them alone has defect at least 0.719, far more than they cost.

With the cap removed, a block of $m$ zeros has defect proportional to $m$, and the remaining question is the energy of eight consecutive zeros. This is an inequality in seven variables, which we verify by branch and bound in outward-rounded arithmetic over a table of interval enclosures of the kernel (Proposition 11 and Appendix B). The extremal configurations are close to a lattice with gaps near 1.04 and 1.98, because the overlap kernel vanishes at 1.0573, 2.0301 and 3.0202.

The method has a limit. Periodic arrangements of zeros satisfy every constraint the argument uses, and their energy shows that no choice of window length can push the bound past 0.67357 (Sec. IX). This is well below the ceiling 0.6818 that Alpöge and Furman identify for all certificates based on pair correlation with Fourier support in $[-1,1]$ [2, §7.2].

The route to this argument was found by the learn-from-failure protocol of Anaxia, the BLLM engine [6, 8].[4] For other approaches to the zeros through spectral and large-sieve methods, see [5, 7].

The paper is organised as follows. Section II recalls the framework of [2]. Section III proves the stability form of the rank–trace inequality and the pinching inequality. Section IV computes the overlap kernel, and Sec. V proves the eigenvalue bound. Section VI bounds the defect of a block of zeros, Sec. VII states the eight-point inequality, and Sec. VIII proves Theorem 1. Section IX shows where the method stops. Appendix A gives the elementary estimates in detail, and Appendix B describes the verification.

## II. THE FRAMEWORK

We use the notation of Alpöge and Furman [2]. Let

$$L := \log\frac{T}{2\pi}, \qquad h := \frac{2\pi}{L},$$
$$I := [T,2T), \qquad I' := [T-\sqrt{T}, 2T+\sqrt{T}),$$

and let $\psi(t) = \cos(\sqrt{2}t)\,\mathbf{1}_{[-1/2,1/2]}(t)$ be the Montgomery–Taylor window. Fix a smooth nondecreasing $\chi$ with $\chi = 0$ on $(-\infty,0]$ and $\chi = 1$ on $[1,\infty)$, and set

$$\phi(u) := \chi\big(\tfrac{L}{2}+u\big)\chi\big(\tfrac{L}{2}-u\big)\,\psi(u/L)^{1/2}, \qquad a := \frac{\|\phi\|_2^2}{L}.$$

With $\widehat{f}(\xi) = \int f(u)e^{-iu\xi}\,du$, the grid points $\alpha_k := T + kh$, and $d := \lfloor LT/2\pi \rfloor$, each zero $\rho$ gives a vector

$$v_\rho := \big(\widehat{\phi}(\gamma_\rho - \alpha_k)\big)_{0\leq k<d}, \qquad \tilde{v}_\rho := \frac{v_\rho}{\sqrt{a}L}.$$

The matrix $\widetilde{G} := \sum_\rho m_\rho \tilde{v}_\rho \tilde{v}_\rho^{\mathsf{T}}$, the sum over the zeros with real part of $\gamma_\rho$ in $I'$, is the compression of Weil's explicit formula to the modulated copies of $\phi$.[5] Let $s_1$ be the number of simple zeros on the line, $s_2$ the number of distinct multiple zeros on the line, and $p$ the number of pairs $\{\rho, 1-\bar{\rho}\}$ off the line, all with real part of the ordinate in $I'$. Let $V$ be the $d \times s_1$ matrix whose columns are the vectors $\tilde{v}_\rho$ of the simple zeros on the line, and put

$$P_1 := VV^{\mathsf{T}}, \qquad Q' := \widetilde{G} - P_1, \qquad M := V^{\mathsf{T}}V.$$

**Proposition 2** ([2]). *As $T \to \infty$:*

*(a) every column of $V$ has norm at most 1, $n_+(Q') \leq s_2 + p$, and $s_1 + 2s_2 + 2p \leq N(I')$;*

*(b) $\operatorname{tr}\widetilde{G} = N(T,2T) + o(N(T,2T))$;*

[4] The protocol records each attempt that fails together with the rule it teaches, and requires every later attempt to respect the rules so far. Eight rounds preceded this paper. The direct attacks on the Riemann hypothesis failed and were recorded as such: a correlation bound below the square-root size of each shift, a Nyman–Beurling completeness argument that uses only integer dilations, and shift-by-shift Chowla bounds. Two rules then shaped the present work. The first was that information beyond pair correlation can be used only where off-line zeros are provably rare; three-point correlations raise the ceiling of the method to about 0.709, but with the zero-density estimates now known only a gain of 0.0002 survives, which closed that route. The second was that a capped block defect saturates near 0.67304, which pointed to the uncapped defect and to the eigenvalue bound of Lemma 8. The engine did not produce the proofs; the failure log, the rules and every computation are in the archive of this paper (see the data availability section).

[5] Weil showed that the Riemann hypothesis is equivalent to the positivity of the Hermitian form $W(f,g) = \sum_\rho m_\rho \widehat{f}(\gamma_\rho)\overline{\widehat{g}(\gamma_\rho)}$ on compactly supported test functions [22]. Off the line the ordinates $\gamma_\rho$ are complex, and a pair of zeros $\rho, 1-\bar{\rho}$ contributes an indefinite form of signature $(1,1)$. The argument of [2] needs no positivity: it counts the positive directions instead.

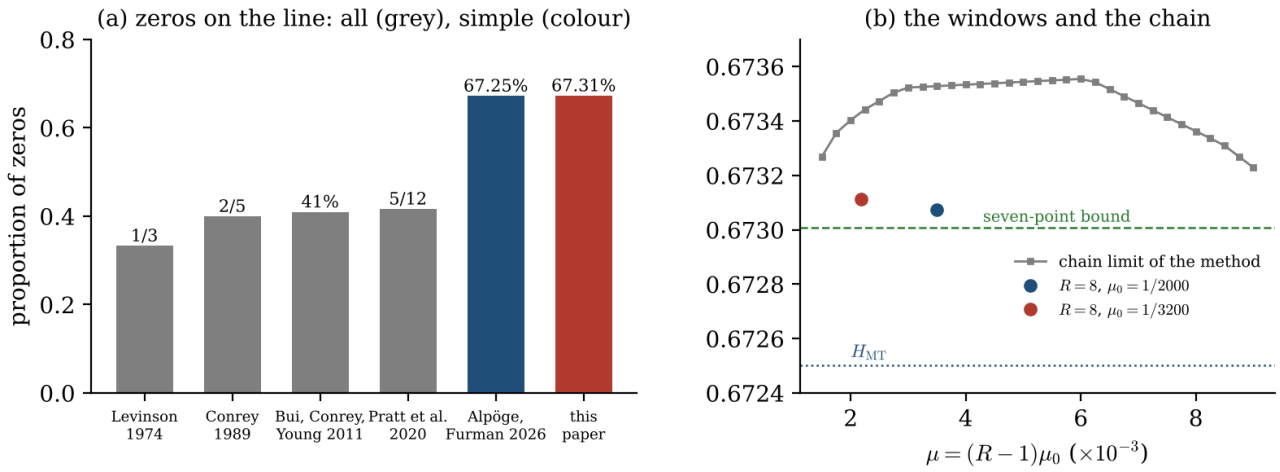



FIG. 1. (a) Unconditional lower bounds for the proportion of zeros on the critical line. The grey bars are the proportions of zeros on the line proved by mollifiers, in the form stated in the titles of the papers; the coloured bars are proportions of zeros that are both simple and on the line. (b) The bound $(H_{\mathrm{MT}}-\mu)/(1-c_0)$ from an inequality for $R$ consecutive zeros, against $\mu=(R-1)\mu_0$: the two certified points for $R=8$, the seven-point bound of [1], the Montgomery–Taylor constant, and the upper limit of the method obtained from periodic configurations (Sec. IX).

*(c)* $\|\widetilde{G}\|_{\mathrm{HS}}^2=(2-H_{\mathrm{MT}}+o(1))N(T,2T)$;

*(d) for all* $z,z'\in\mathbb{C}$, $\sum_{k\in\mathbb{Z}}\widehat{\phi}(z-\alpha_k)\widehat{\phi}(z'-\alpha_k)=L\widehat{\phi^2}(z-z')$, *and* $|\widehat{\phi}(r)|\ll\min(L,r^{-2})$ *for real* $r$.

*Proof.* Part (a) is Lemma 2.1 at $z=z'=\gamma_\rho$, Proposition 4.1 and the decomposition in the proof of Theorem A in [2]. Part (b) is Proposition 4.2 there together with $N(I')-N(I)\ll\sqrt{T}\log T$. Part (c) is Theorem 5.7 with Lemma 5.6, since $R(\psi)=2-H_{\mathrm{MT}}$ for the Montgomery–Taylor window. Part (d) is Lemma 2.1 and the bound (2.8) of [2]. □

Here $n_+$ is the number of positive eigenvalues and $\|X\|_{\mathrm{HS}}^2=\operatorname{tr}X^2$. Proposition 2(b) and (c) are where the prime numbers enter: (c) is Montgomery's second moment, evaluated without the Riemann hypothesis [3].

## III. THE STABILITY DEFECT

Define the convex function

$$\Psi(t):=\begin{cases}(t-1)^2, & 0\le t\le 2,\\ 2t-3, & t\ge 2,\end{cases}$$

and, for a positive semidefinite matrix $X$, the stability defect $\mathscr{D}(X):=\operatorname{tr}\Psi(X)\ge 0$. The function $\Psi$ is continuously differentiable, with $\Psi'(2)=2$ from both sides.

**Lemma 3** (Rank–trace inequality with defect)**.** *Let $V$ be a $d\times r$ matrix whose columns have norm at most 1, put $P=VV^*$ and $M=V^*V$, and let $Q$ be Hermitian with $n_+(Q)\le b$. Then*

$$\|P+Q\|_{\mathrm{HS}}^2\ge 4\operatorname{tr}(P+Q)-3r-4b+\mathscr{D}(M).$$

The case $\mathscr{D}(M)=0$ is the rank–trace inequality of [2, Lemma 3.2]; the defect term is the refinement of [1]. We give the proof.

*Proof.* Write $Q=Q_+-Q_-$ with $Q_\pm\succeq 0$, $Q_+Q_-=0$ and $\operatorname{rank}Q_+\le b$. Since $\operatorname{tr}(PQ_+)\ge 0$ and $\operatorname{tr}(Q_+Q_-)=0$,

$$\|P+Q\|_{\mathrm{HS}}^2\ge\|P-Q_-\|_{\mathrm{HS}}^2+\|Q_+\|_{\mathrm{HS}}^2.$$

Every positive eigenvalue $q$ of $Q_+$ satisfies $q^2\ge 4q-4$, so $\|Q_+\|_{\mathrm{HS}}^2\ge 4\operatorname{tr}Q_+-4b$. Let $p_1\ge\cdots\ge p_d\ge 0$ be the eigenvalues of $P$; the first $r$ of them are the eigenvalues of $M$ and the others vanish. Let $n_1\ge\cdots\ge n_d\ge 0$ be the eigenvalues of $Q_-$. Von Neumann's trace inequality $\operatorname{tr}(PQ_-)\le\sum_i p_in_i$ gives

$$\|P-Q_-\|_{\mathrm{HS}}^2+4\operatorname{tr}Q_-\ge\sum_{i\le r}\bigl((p_i-n_i)^2+4n_i\bigr),$$

because the terms with $i>r$ are $n_i^2+4n_i\ge 0$. For $p\ge 0$,

$$\min_{n\ge 0}\bigl((p-n)^2+4n\bigr)=2p-1+\Psi(p),$$

the minimum being at $n=0$ when $p\le 2$ and at $n=p-2$ when $p\ge 2$.[6] Hence $\|P-Q_-\|_{\mathrm{HS}}^2\ge 2\operatorname{tr}P-r+\mathscr{D}(M)-4\operatorname{tr}Q_-$. Adding the two bounds,

$$\|P+Q\|_{\mathrm{HS}}^2\ge 4\operatorname{tr}(P+Q)-2\operatorname{tr}P-r-4b+\mathscr{D}(M),$$

and $\operatorname{tr}P=\sum\|v_j\|^2\le r$. □

**Lemma 4** (Pinching)**.** *Let $X\succeq 0$ be an $n\times n$ matrix and let $B_1,\dots,B_q$ be a partition of $\{1,\dots,n\}$. Then $\mathscr{D}(X)\ge\sum_j\mathscr{D}(X_{B_jB_j})$, where $X_{BB}$ is the principal submatrix on $B$.*

[6] For $p\le 2$ the minimum is $p^2=2p-1+(p-1)^2$, and for $p\ge 2$ it is $4+4(p-2)=4p-4=2p-1+(2p-3)$. The function $\Psi$ is thus the exact loss in the scalar inequality $p^2\ge 2p-1$ when one allows a compensating negative eigenvalue; this is what makes the defect the right quantity to keep.

*Proof.* Let $\Pi_j$ be the coordinate projection onto $B_j$ and, for $\varepsilon \in \{\pm 1\}^q$, let $U_\varepsilon = \sum_j \varepsilon_j \Pi_j$. Then $\sum_j \Pi_j X \Pi_j = 2^{-q} \sum_\varepsilon U_\varepsilon X U_\varepsilon$. The map $Y \mapsto \operatorname{tr}\Psi(Y)$ is convex on positive semidefinite matrices, since $\Psi$ is convex on $[0,\infty)$ [4, Theorem V.2.5], and it is unitarily invariant. Hence $\mathscr{D}(\sum_j \Pi_j X \Pi_j) \le \mathscr{D}(X)$, and the left side is $\sum_j \mathscr{D}(X_{B_j B_j})$. □

**Corollary 5.** *Let $M^\circ$ be the principal submatrix of $M$ indexed by any set of simple zeros on the line with ordinates in $I'$. Then*

$$N_0^s(T,2T) \ge H_{\mathrm{MT}} N(T,2T) + \mathscr{D}(M^\circ) - o(N(T,2T)).$$

*Proof.* Apply Lemma 3 to $P_1 + Q' = \widetilde{G}$ with $r = s_1$ and $b = s_2 + p$ (Proposition 2(a)):

$$\begin{aligned} \|\widetilde{G}\|_{\mathrm{HS}}^2 &\ge 4 \operatorname{tr}\widetilde{G} - 3s_1 - 4s_2 - 4p + \mathscr{D}(M) \\ &\ge 4 \operatorname{tr}\widetilde{G} - s_1 - 2N(I') + \mathscr{D}(M). \end{aligned}$$

By Proposition 2(b), (c) and $N(I') = N(T,2T) + o(N(T,2T))$, this is $s_1 \ge H_{\mathrm{MT}} N(T,2T) + \mathscr{D}(M) - o(N(T,2T))$. Lemma 4, applied to $M^\circ$ and its complement, gives $\mathscr{D}(M) \ge \mathscr{D}(M^\circ)$, and $N_0^s(T,2T) \ge s_1 - O(\sqrt{T}\log T)$. □

## IV. THE OVERLAP KERNEL

Put

$$K(x) := \int_{-1/2}^{1/2} \cos(\sqrt{2}t)\cos(2\pi x t)\,dt,$$
$$k(x) := \frac{K(x)}{K(0)}, \qquad w := k^2,$$

so that $K(0) = K_0 := \sqrt{2}\sin(1/\sqrt{2}) = 0.9187253698\ldots$ and

$$K(x) = \frac{1}{2}\Big(\operatorname{sinc}\big(\pi x - \tfrac{1}{\sqrt{2}}\big) + \operatorname{sinc}\big(\pi x + \tfrac{1}{\sqrt{2}}\big)\Big), \tag{2}$$

with $\operatorname{sinc} s = (\sin s)/s$. For a simple zero $\rho$ on the line put $x_\rho := (\gamma_\rho - T)/h$, its ordinate measured in units of $h$, which is the mean spacing of the zeros up to a factor $1 + O(1/L)$.

**Lemma 6.** *Retain the simple zeros on the line with $x_\rho \in [L^2, d - L^2]$. The others number $o(N(T,2T))$ among those in $I$, and for every fixed $S > 0$, uniformly for retained zeros with $|x_\rho - x_{\rho'}| \le S$,*

$$\langle \tilde{v}_\rho, \tilde{v}_{\rho'} \rangle = k(x_\rho - x_{\rho'}) + o(1).$$

*Proof.* By Proposition 2(d), the sum over all $k \in \mathbb{Z}$ equals $(aL)^{-1}\widehat{\phi^2}(hx)$ with $x = x_\rho - x_{\rho'}$. The grid points outside $[0,d)$ lie at distance at least $hL^2 = 2\pi L$ from $\gamma_\rho$ and $\gamma_{\rho'}$, so by Cauchy–Schwarz and the decay of $\widehat{\phi}$ they contribute $\ll L^{-2}\sum_{j\ge 0}(2\pi L + jh)^{-4} \ll L^{-4}$ after division by $aL^2$. For the full sum, the substitution $u = Lt$ gives

$$\frac{\widehat{\phi^2}(hx)}{aL} = \frac{\int \phi(Lt)^2 e^{-2\pi i x t}\,dt}{\int \phi(Lt)^2\,dt}.$$

As $T \to \infty$, $\phi(Lt)^2 \to \psi(t)$ in $L^1$, because the two factors $\chi$ differ from 1 only on a set of $t$ of measure $O(1/L)$. The ratio therefore tends to $K(x)/K_0$, uniformly for $|x| \le S$. The deleted zeros have ordinates in two intervals of length $2\pi L$ at the ends of $I$, and there are $O(L^2)$ of them by $N(t,t+1) \ll \log t$ [20, Theorem 9.2]. □

The kernel has three properties that the argument uses (Fig. 2).

**Lemma 7.** *(i) $k(0) = 1$, $|k| \le 1$, and $k$ is decreasing on $[0,1]$.*

*(ii) $w(x) \ge w(\frac{11}{20}) = 0.3595640596\ldots$ for $0 \le x \le \frac{11}{20}$.*

*(iii) The first three positive zeros of $k$ are $1.0572782\ldots$, $2.0300675\ldots$ and $3.0202429\ldots$, and $w(x) = O(x^{-2})$.*

*Proof.* (i) $|K(x)| \le \int \cos(\sqrt{2}t)\,dt = K_0$. For $0 < x \le 1$ and $0 < t \le \frac{1}{2}$ one has $0 < 2\pi x t \le \pi$, so

$$K'(x) = -\int_{-1/2}^{1/2} 2\pi t \cos(\sqrt{2}t)\sin(2\pi x t)\,dt < 0.$$

(ii) follows from (i) and $k(\frac{11}{20}) > 0$; the value is computed in Appendix A. (iii) The zeros are those of (2); the decay follows from the same formula. □

## V. SEPARATED ZEROS

**Lemma 8.** *Let $\delta_1 = \frac{11}{20}$ and let $x_1,\ldots,x_n$ be real numbers with $|x_i - x_j| \ge \delta_1$ for $i \ne j$. Then every eigenvalue of the matrix $\big(k(x_i - x_j)\big)_{i,j}$ lies in*

$$\Big[0,\ \frac{1}{\delta_1 K_0}\Big] = [0,\ 1.9790264\ldots].$$

*Proof.* For a real vector $c$, with $S(t) = \sum_j c_j e^{2\pi i x_j t}$,

$$\sum_{i,j} c_i c_j k(x_i - x_j) = \frac{1}{K_0}\int_{-1/2}^{1/2} \cos(\sqrt{2}t)\,|S(t)|^2\,dt \ge 0,$$

because the cosine is positive on $[-\frac{1}{2},\frac{1}{2}]$. Let $\Lambda(t) = \operatorname{sinc}^2(\pi\delta_1 t)$. Its Fourier transform is the triangle

$$\int_{\mathbb{R}} \Lambda(t) e^{2\pi i \xi t}\,dt = \frac{1}{\delta_1}\Big(1 - \frac{|\xi|}{\delta_1}\Big)_+,$$

which vanishes for $|\xi| \ge \delta_1$. We show in Appendix A that

$$\Lambda(t) \ge \cos(\sqrt{2}t) \qquad \text{for } |t| \le \tfrac{1}{2}. \tag{3}$$

Since $\Lambda \ge 0$ everywhere, $\Lambda$ majorises $\cos(\sqrt{2}t)\mathbf{1}_{[-1/2,1/2]}$ on the whole line, so

$$\begin{aligned} \int_{-1/2}^{1/2} \cos(\sqrt{2}t)|S|^2\,dt &\le \int_{\mathbb{R}} \Lambda |S|^2\,dt \\ &= \sum_{i,j} c_i c_j \widehat{\Lambda}(x_i - x_j) = \frac{\|c\|^2}{\delta_1}, \end{aligned}$$

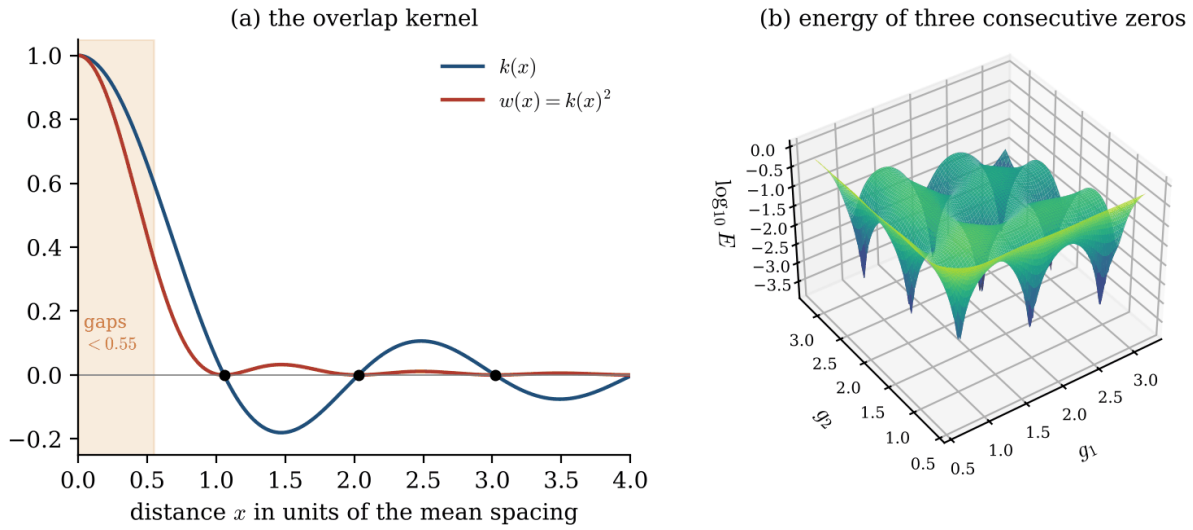



FIG. 2. (a) The overlap kernel $k$ and its square $w$, in units of the mean spacing; the dots are the zeros of $k$, near 1, 2 and 3. The shaded strip marks the gaps below $\frac{11}{20}$, where the eigenvalue bound of Lemma 8 no longer applies and zeros are treated as clusters. (b) The energy $E = 2[w(g_1)+w(g_2)+w(g_1+g_2)]$ of three consecutive zeros with gaps $g_1, g_2$, on a logarithmic scale. Its valleys follow the zeros of $k$; the extremal configurations of Proposition 11 sit in these valleys, with gaps near 1.04 and 1.98.

because $\widehat{\Lambda}(x_i - x_j) = 0$ for $i \neq j$.[7] Dividing by $K_0$ gives the upper bound. □

The constant is sharp. For the points $x_j = j\delta$, $1 \le j \le n$, and the vector of ones, the Rayleigh quotient is $\sum_{|\ell|<n}(1 - |\ell|/n)\,k(\ell\delta)$, the $n$-th Fejér mean at 0 of the Fourier series of the 1-periodic function $\theta \mapsto (\delta K_0)^{-1}\sum_{j\in\mathbb{Z}} \psi((j-\theta)/\delta)$ by Poisson summation. For $\delta < 2$ only the term $j = 0$ is nonzero near $\theta = 0$, the function is continuous there with value $1/(\delta K_0)$, and Fejér's theorem gives $\lambda_{\max} \ge 1/(\delta K_0) - o(1)$ as $n \to \infty$. So lattices with spacing below the critical value

$$\delta^* = \frac{1}{2K_0} = 0.5442322\ldots$$

have eigenvalues above 2, and Lemma 8 cannot hold with $\frac{1}{2}$ in place of $\frac{11}{20}$. Figure 3 shows the majorant and the eigenvalues of lattices.

[7] This is the mechanism of the large sieve: a band-limited majorant of the weight turns a quadratic form over separated points into its diagonal. Beurling and Selberg chose the majorant to optimise the large sieve constant; see [21]. Here the weight is $\cos(\sqrt{2}t)$ on an interval of length 1 rather than the indicator, and the separation $\delta_1$ is below 1, the range in which the classical constant $1 + 1/\delta$ exceeds 2 and is useless for us. The Fejér kernel $\Lambda$ with exactly the bandwidth $\delta_1$ wins because $\cos(\sqrt{2}t)$ is flat to second order at $t = 0$, where $\Lambda$ has curvature $-(\pi\delta_1)^2/3 \approx -0.995$, almost the $-1$ of the cosine.

## VI. THE DEFECT OF A BLOCK OF ZEROS

Fix an integer $R \ge 2$ and $\mu_0 > 0$. For gaps $g = (g_1, \ldots, g_{R-1})$ in $[0,\infty)^{R-1}$ put $y_0 = 0$, $y_i = g_1 + \cdots + g_i$, and

$$F_R(g) := \mu_0 \sum_{i=1}^{R-1} g_i + \sum_{r=1}^{R-1} \frac{2}{R-r} \sum_{i=0}^{R-1-r} w(y_{i+r} - y_i). \qquad (4)$$

The second sum runs over all pairs of the $R$ points $y_0 < \cdots < y_{R-1}$; a pair $r$ steps apart carries the weight $2/(R-r)$.

**Lemma 9.** *Suppose that $F_R \ge c_0$ on $[0,\infty)^{R-1}$ and put $\mu = (R-1)\mu_0$. Then for all real $y_1 < \cdots < y_n$,*

$$2\sum_{i<j} w(y_j - y_i) + \mu\,(y_n - y_1) \ge c_0\,(n - R + 1).$$

*Proof.* For $n < R$ the right side is not positive. For $n \ge R$, add the inequality $F_R \ge c_0$ over the $n - R + 1$ windows of $R$ consecutive points. A pair of points $r < R$ steps apart lies in at most $R - r$ windows, with weight $2/(R-r)$ each time, so its total weight is at most 2; pairs at least $R$ steps apart do not occur and $w \ge 0$. Each gap lies in at most $R - 1$ windows, so the linear terms add up to at most $\mu(y_n - y_1)$. □

For a set $B$ of retained simple zeros, let $G_B$ be its Gram matrix $(\langle \tilde{v}_\rho, \tilde{v}_{\rho'} \rangle)_{\rho,\rho' \in B}$ and let $\operatorname{span}(B)$ be the difference between the largest and the smallest $x_\rho$ in $B$.

**Proposition 10.** *Suppose that $F_R \ge c_0$ on $[0,\infty)^{R-1}$ with*

$$(R+1)\,c_0 \le \tfrac{4}{3}\,w(\tfrac{11}{20}), \qquad (5)$$

*and let $\mu = (R-1)\mu_0$. Fix $m \ge R$. Then, uniformly over all sets $B$ of $m$ consecutive retained simple zeros on the line,*

$$\mathscr{D}(G_B) + \mu \operatorname{span}(B) \ge c_0\,(m - R + 1) - o(1).$$

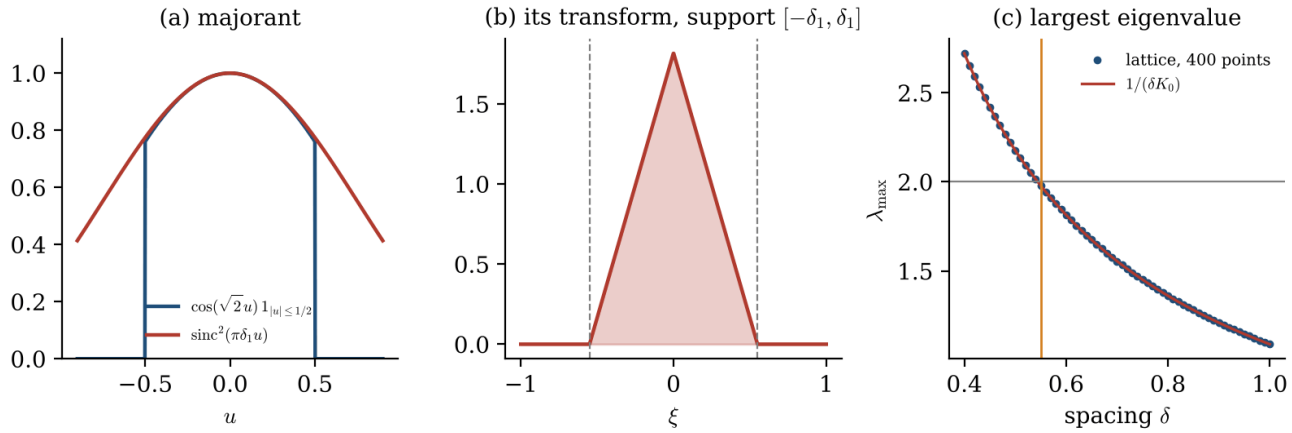



FIG. 3. The eigenvalue bound. (a) The Fejér kernel $\mathrm{sinc}^2(\pi\delta_1 u)$ lies above the Montgomery–Taylor window on $[-\frac{1}{2},\frac{1}{2}]$; the two curves agree to second order at $u=0$. (b) Its Fourier transform, a triangle supported in $[-\delta_1,\delta_1]$, so that points at distance at least $\delta_1$ do not interact. (c) The largest eigenvalue of the overlap matrix of 400 evenly spaced points against the spacing $\delta$, with the bound $1/(\delta K_0)$; the vertical line is $\delta_1 = \frac{11}{20}$ and the horizontal line is the threshold 2 of the defect.

*Proof.* If $\mu\,\mathrm{span}(B) \ge c_0(m-R+1)$ there is nothing to prove, since $\mathscr{D} \ge 0$. Otherwise $\mathrm{span}(B) < S := c_0(m-R+1)/\mu$, so by Lemma 6 every entry of $G_B$ is $k(x_\rho - x_{\rho'}) + o(1)$, uniformly in $B$. Write $y_1 < \cdots < y_m$ for the points $x_\rho$, $\rho \in B$.

Call a gap $y_{i+1} - y_i$ short if it is less than $\delta_1 = \frac{11}{20}$. A cluster is a maximal set of at least two consecutive points joined by short gaps; let there be $q$ clusters, of sizes $s_1,\dots,s_q \ge 2$. The points outside the clusters fall into at most $q+1$ maximal sets of consecutive points, the runs, of sizes $n_1, n_2, \dots$; inside a run every gap is at least $\delta_1$, so all distances are. Split each cluster into $\lfloor s_j/2 \rfloor$ pairs of consecutive points and, when $s_j$ is odd, one single point. Runs, pairs and single points partition $B$ (Fig. 4), and Lemma 4 gives

$$\mathscr{D}(G_B) \ge \sum_{\text{runs}} \mathscr{D}(G_{\text{run}}) + \sum_{\text{pairs}} \mathscr{D}(G_{\text{pair}}).$$

The function $X \mapsto \mathscr{D}(X)$ is uniformly continuous on the matrices of size at most $m$ with entries bounded by 2, so in each term we may replace $G$ by the limiting matrix $(k(y_i - y_j))$ at a cost $o(1)$. For a run, the limiting matrix has unit diagonal and, by Lemma 8, all its eigenvalues in $[0,2]$; hence its defect is $\mathrm{tr}(X-I)^2$, which is twice the sum of $w$ over the pairs of points of the run. For a pair at distance $g < \delta_1$ the limiting matrix has eigenvalues $1 \pm k(g) \in [0,2]$, and its defect is $2w(g) \ge 2w(\delta_1)$ by Lemma 7(ii). With Lemma 9 for each run and $\mathrm{span}(B) \ge \sum \mathrm{span}(\text{run})$,

$$\begin{aligned}
\mathscr{D}(G_B) + \mu\,\mathrm{span}(B) &\ge c_0 \sum_{\text{runs}} (n-R+1) - o(1) \\
&\quad + 2w(\delta_1) \sum_j \Big\lfloor \frac{s_j}{2} \Big\rfloor \\
&\ge c_0(m-R+1) - o(1) \\
&\quad + \sum_{j=1}^{q} \Big( 2w(\delta_1) \Big\lfloor \frac{s_j}{2} \Big\rfloor - c_0 s_j \\
&\qquad - c_0(R-1) \Big),
\end{aligned}$$

using $\sum n + \sum s_j = m$ and at most $q+1$ runs. Since $\lfloor s/2 \rfloor \ge s/3$ for $s \ge 2$, and $2w(\delta_1)/3 > c_0$ by (5), each bracket is at least $s_j(\frac{2}{3}w(\delta_1) - c_0) - c_0(R-1) \ge \frac{4}{3}w(\delta_1) - (R+1)c_0 \ge 0$. □

## VII. EIGHT CONSECUTIVE ZEROS

**Proposition 11.** *With $R = 8$ and $\mu_0 = \frac{1}{3200}$,*

$$F_8(g) \ge \frac{13}{3125} = 0.00416 \qquad \textit{for all } g \in [0,\infty)^7.$$

The inequality is verified by a computation in rigorous arithmetic, described in full in Appendix B. The search covers the whole of $[0,\infty)^7$: if $\sum g_i \ge 13.32$ the linear term alone exceeds 0.00416, and on the remaining simplex the domain is subdivided into boxes of grid cells of width $\frac{1}{4000}$ until a lower bound computed from interval enclosures of $w$ settles each box. A local minimisation finds the smallest value 0.0041773, at the gaps

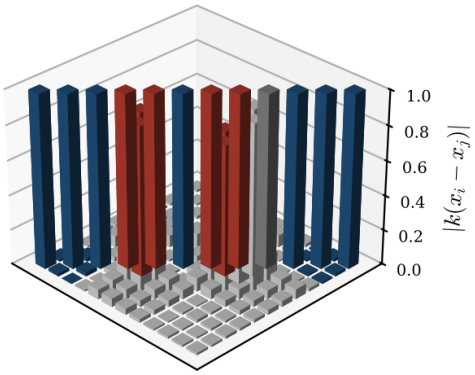
$$(1.0461,\ 1.9894,\ 1.9868,\ 1.0419,\ 1.9868,\ 1.9894,\ 1.0461),$$

so the margin is 0.4 percent.[8] The same search also certifies $F_8 \ge \frac{605}{100000}$ for $\mu_0 = \frac{1}{2000}$, which gives the slightly weaker bound $0.6730727\ldots$ by the argument of Sec. VIII.

The inequality of [1] is the case $R = 7$, $\mu_0 = \frac{1}{3000}$, $c_0 = \frac{19}{5000}$. There the defect of a whole block was capped at 1, so blocks could contain at most about $1/c_0$ windows, and a longer window gains nothing. Lemma 8 removes the cap, so the size of the blocks can tend to infinity and the window length can be chosen freely.

[8] The extremal configuration is a piece of a near-lattice: gaps of about one and two mean spacings, each slightly larger than the zero of $k$ near 1 or below the zero near 2, so that most overlaps vanish and the remaining ones balance the linear term. Periodic arrangements of the same two gaps are the extremal chains of Sec. IX for the weights near the one used here.

(a) Gram matrix; the kept blocks in colour




(b) runs, pairs and single zeros

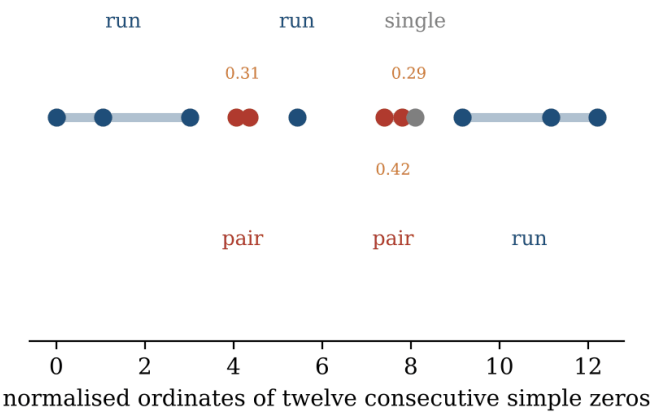



FIG. 4. The decomposition in the proof of Proposition 10, for twelve consecutive zeros with three short gaps (orange). (a) The Gram matrix; pinching keeps the coloured diagonal blocks and discards the grey entries. (b) The runs have all gaps at least $\frac{11}{20}$, so their eigenvalues stay below 2 and their defect is their full energy; the pairs joined by a short gap have defect at least 0.719 each, which pays for the windows that the clusters cut.

## VIII. PROOF OF THEOREM 1

Take $R = 8$, $\mu_0 = \frac{1}{3200}$, $c_0 = \frac{13}{3125}$, so that $\mu = \frac{7}{3200}$. Condition (5) holds, since $9c_0 = 0.03744 < 0.4794 < \frac{4}{3}w(\frac{11}{20})$. Fix $m \geq 8$ and let $x_1 < \cdots < x_{S^\circ}$ be the normalised ordinates of the retained simple zeros on the line, so that $S^\circ = N_0^s(T,2T) - o(N(T,2T))$ by Lemma 6. Let $M^\circ$ be their Gram matrix.

For each offset $j \in \{0,\dots,m-1\}$, partition the points into the first $j$ of them, consecutive blocks of $m$ points, and a final piece of fewer than $m$ points. By Lemma 4 and Proposition 10,

$$\mathscr{D}(M^\circ) \geq \sum_B \mathscr{D}(G_B) \geq \sum_B \big(c_0(m-R+1) - \mu\,\mathrm{span}(B) - o(1)\big).$$

For every offset there are at least $S^\circ/m - 2$ blocks and at most $S^\circ/m$, and the $o(1)$ is uniform, so these terms total $o(N)$, with $N = N(T,2T)$. Average over the $m$ offsets. A gap $x_{i+1} - x_i$ lies inside a block for at most $m-1$ of the $m$ offsets, and $x_{S^\circ} - x_1 \leq d = N(1 + O(1/L))$, so

$$\mathscr{D}(M^\circ) \geq \frac{c_0(m-R+1)}{m}\,S^\circ - \frac{(m-1)\mu}{m}\,N - o(N).$$

Insert this in Corollary 5:

$$\Big(1 - \frac{c_0(m-R+1)}{m}\Big)N_0^s(T,2T) \geq \Big(H_{\mathrm{MT}} - \frac{(m-1)\mu}{m}\Big)N - o(N).$$

Divide by $N$ and let $T \to \infty$, and then let $m \to \infty$:

$$\liminf_{T\to\infty} \frac{N_0^s(T,2T)}{N(T,2T)} \geq \frac{H_{\mathrm{MT}} - \mu}{1 - c_0} = \frac{3125}{3112}\Big(H_{\mathrm{MT}} - \frac{7}{3200}\Big).$$

The value is 0.6731133552371983…, computed in ball arithmetic [13]. □

*Remark* 12. Every input of the proof is available for a Dirichlet $L$-function $L(s,\chi)$ with $\chi$ primitive: Proposition 2 holds there by [2, Theorem B], and Lemmas 3–9 do not involve $\zeta$. The same constant therefore holds for the simple zeros of $L(s,\chi)$ on the critical line.

## IX. HOW FAR THE METHOD GOES

The argument has three parameters: the window length $R$, the weight $\mu_0$, and the constant $c_0$, which is the minimum of $F_R$. Longer windows see more of the energy, and the local minima suggest that $R = 9$ with $\mu_0 = \frac{1}{2400}$ would give about 0.67317. The following proposition bounds what any $R$ can give.

**Proposition 13.** *Let $(g_i)_{i\in\mathbb{Z}}$ be a periodic sequence of positive gaps with period $p$ and mean $\bar{g}$, let $y$ be the corresponding points, and put*

$$e(g,\mu) = \mu\bar{g} + \frac{2}{p}\sum_{i=1}^{p}\sum_{j>i} w(y_j - y_i).$$

*If $F_R \geq c_0$ holds with $\mu = (R-1)\mu_0$, then $c_0 \leq e(g,\mu)$, and the bound obtained in Sec. VIII is at most $(H_{\mathrm{MT}} - \mu)/(1 - e(g,\mu))$.*

*Proof.* Apply Lemma 9 to $n$ consecutive points of the sequence, divide by $n$ and let $n \to \infty$; the double sum converges because $w(x) = O(x^{-2})$. □

TABLE I. The chain energy $e(\mu)$, the smallest value of $e(g,\mu)$ found over periodic gap patterns $g$ of period at most 4, the pattern that attains it, and the resulting upper limit $(H_{\mathrm{MT}}-\mu)/(1-e(\mu))$ for the bound of Sec. VIII.

| $\mu$ | $e(\mu)$ | pattern | upper limit |
|---|---|---|---|
| 0.0020 | 0.0043102 | $(1.99, 1.98, 1.04, 1.98)$ | 0.6734032 |
| 0.0025 | 0.0051570 | $(1.98, 1.04, 1.98)$ | 0.6734738 |
| 0.0030 | 0.0059725 | $(1.04, 1.96)$ | 0.6735233 |
| 0.0040 | 0.0074727 | $(1.04, 1.96)$ | 0.6735338 |
| 0.0050 | 0.0089729 | $(1.04, 1.96)$ | 0.6735443 |
| 0.0060 | 0.0104731 | $(1.04, 1.96)$ | 0.6735549 |
| 0.0065 | 0.0111605 | $(1.02, 1.02, 1.95)$ | 0.6735175 |
| 0.0080 | 0.0131609 | $(1.02, 1.02, 1.95)$ | 0.6733628 |

Minimising $e(g,\mu)$ over periodic patterns of period at most 4 on a grid of weights gives the values in Table I. For a fixed pattern the upper limit $(H_{\mathrm{MT}}-\mu)/(1-e(g,\mu))$ is monotone in $\mu$, since its derivative has the sign of $\bar{g}H_{\mathrm{MT}}+e(g,0)-1$, so the patterns found at the grid points bound it on every interval between them.[9] The resulting bound stays below 0.67357 for every $\mu \geq 0$, with the largest value near $\mu = 0.0061$, where the best pattern is the alternation $(1.036, 1.964)$; for smaller $\mu$ gaps near 2 take over, and for larger $\mu$ patterns with more gaps near 1. So no window length improves Theorem 1 by more than 0.0005, and the approach cannot reach 67.36 percent. Tilting the window, that is replacing $\cos(\sqrt{2}t)$ by $\cos(\alpha t)$, lowers $H_{\mathrm{MT}}$ faster than it raises the defect: at $R = 7$ the best tilt, $\alpha = 1.5$, gains 0.00004.

Going further needs information that the energy of runs does not see. Alpöge and Furman show that no certificate built from pair correlation with Fourier support in $[-1,1]$ can exceed about 0.6818 [2, §7.2]. Between 0.6736 and 0.6818 lies the part of that information which the stability defect, a function of the Gram matrix of the simple zeros alone, does not capture. Information beyond pair correlation can raise the ceiling: in the periodic models of [2, §7.2] with the three-point correlations of the circular unitary ensemble added, the ceiling rises to about 0.709. These correlations are known in the range used, but an off-line pair of zeros enters a triple product with a weight that grows faster than in the quadratic case, and the known zero-density estimates control only the triples whose frequencies add up to less than half of the range. With those alone the gain is about 0.0002.

None of this bears on the Riemann hypothesis itself. Every bound of this kind is insensitive to a set of zeros of density zero, and the argument applies equally to functions, such as the Davenport–Heilbronn function, whose zeros are known to leave the critical line.

[9] The double sum is taken over all pairs: $2\times 10^5$ neighbours are summed and the rest is bounded by $|k(x)| \leq (K_0(\pi x - 1/\sqrt{2}))^{-1}$. A truncation at a few hundred neighbours, which looks harmless because $w$ is small there, lowers $e(g,\mu)$ by about $5\times 10^{-5}$, more than the margins in the table.

## DATA AND CODE AVAILABILITY

The proofs of Lemmas 3–9 and of Proposition 10 are by hand. Proposition 11 is verified by the rigorous computation described in Appendix B. The repository `creelie/mh-phys` contains the LaTeX source of this paper; the table of interval enclosures of $w$ in 128-bit ball arithmetic with its hash; the branch-and-bound verifier, in a serial and a parallel version, with the logs of the certified runs and a workflow that repeats the certificate on every change; independent checks in ball and floating-point arithmetic of the eigenvalue bound, of the endpoint values in Appendix A and of the eigenvalues of lattices; the local minimisations for $R = 7, 8, 9$; the chain energies of Table I and their extension to every weight; the linear programmes behind the ceilings quoted in Sec. IX; and the failure log of the search that led to the argument. The archive is [23], `https://github.com/creelie/mh-phys`, where later versions are listed. The search followed the learn-from-failure protocol of the BLLM engine Anaxia, release 3.0.5 [8]. Deep Bhattacharjee carried out the computations and the C, Python and shell checks with AI assistance from Claude (Anthropic); both authors approved the final text.

## ETHICAL DECLARATIONS

*Funding.* None. *Competing interests.* None.

### Appendix A: Elementary estimates

#### 1. The majorant

We prove (3). Both sides are even, so let $0 \leq t \leq \frac{1}{2}$ and $s = \pi\delta_1 t \in [0, 0.8640]$. The Taylor series

$$\operatorname{sinc}^2 s = \frac{1-\cos 2s}{2s^2} = \sum_{k\geq 1} (-1)^{k+1} \frac{2^{2k-1} s^{2k-2}}{(2k)!}$$

alternates, and for $s \leq 1$ the ratio of consecutive terms is $4s^2/((2k+1)(2k+2)) < 1$, so the terms decrease in absolute value and the partial sum ending with a negative term is a lower bound:

$$\operatorname{sinc}^2 s \geq 1 - \frac{s^2}{3} + \frac{2s^4}{45} - \frac{s^6}{315}.$$

In the same way $\cos y \leq 1 - \frac{y^2}{2} + \frac{y^4}{24}$ for all real $y$. With $y = \sqrt{2}t$ and $u = t^2 \in [0, \frac{1}{4}]$,

$$\Lambda(t) - \cos(\sqrt{2}t) \geq -u\, q(u),$$
$$q(u) = (\alpha_1 - 1) + \left(\tfrac{1}{6} - \alpha_2\right)u + \alpha_3 u^2,$$

where $\alpha_1 = (\pi\delta_1)^2/3 = 0.9951851\ldots$, $\alpha_2 = 2(\pi\delta_1)^4/45 = 0.3961573\ldots$ and $\alpha_3 = (\pi\delta_1)^6/315 = 0.0844821\ldots$. The quadratic $q$ is convex, so its maximum on $[0, \frac{1}{4}]$ is at an endpoint, and

$$q(0) = -0.0048148\ldots < 0, \qquad q(\tfrac{1}{4}) = -0.0569074\ldots < 0.$$

Hence $\Lambda(t) \ge \cos(\sqrt{2}t)$ on $[0, \frac{1}{2}]$. The values are computed with $\pi$ enclosed in 200-bit ball arithmetic; $q(0) = \alpha_1 - 1$ alone needs only $\pi^2 \cdot 0.3025 < 3$, that is $\pi^2 < 9.918$. The margin at small $t$ is of order $0.0048t^2$, which is why $\delta_1$ cannot be lowered much: $q(0) < 0$ requires $\delta_1 < \sqrt{3}/\pi = 0.5513$.

### 2. The Fourier transform of the Fejér kernel

For $\Delta > 0$, the triangle $\tau(\xi) = \Delta^{-1}(1 - |\xi|/\Delta)_+$ is the convolution of $\Delta^{-1}\mathbf{1}_{[-\Delta/2,\Delta/2]}$ with itself, whose inverse Fourier transform is $\operatorname{sinc}(\pi\Delta t)$. Hence $\int \tau(\xi) e^{-2\pi i \xi t}\, d\xi = \operatorname{sinc}^2(\pi\Delta t)$, and Fourier inversion gives the formula used in the proof of Lemma 8.

### 3. Values of the kernel

From (2), in 200-bit ball arithmetic,

$$\begin{aligned} K_0 &= 0.91872536986556843778\ldots, \\ 1/(\delta_1 K_0) &= 1.97902646189890434914\ldots, \\ k(\tfrac{11}{20}) &= 0.59963660635252251069\ldots, \\ w(\tfrac{11}{20}) &= 0.35956405967797003982\ldots, \end{aligned}$$

so that $\frac{4}{3}w(\frac{11}{20}) = 0.4794187\ldots$. The minimum of $k$ on $[0,\infty)$ is $-0.1804\ldots$, attained near $x = 1.47$.

### 4. Convex trace functions

If $f$ is convex on an interval $J$, then $X \mapsto \operatorname{tr} f(X)$ is convex on the Hermitian matrices with spectrum in $J$ [4, Theorem V.2.5]. For the pinching we need it on positive semidefinite matrices, with $f = \Psi$; the averages $2^{-q}\sum_\varepsilon U_\varepsilon X U_\varepsilon$ of Lemma 4 remain positive semidefinite, so the spectrum stays in $[0,\infty)$, where $\Psi$ is convex.

## Appendix B: Verification of Proposition 11

### 1. The table

For $c = 0, 1, \ldots, 45999$, a lower bound $\underline{w}_c$ for $w$ on the cell $[c/4000, (c+1)/4000]$ is computed by evaluating (2) on the cell as a ball in 128-bit arithmetic [13], taking the lower bound of $|k|$, rounding it down to a double, squaring, and rounding down again. On $[11.5,\infty)$ only $w \ge 0$ is used. The table occupies 368,000 bytes and has SHA-256 hash `8ea9e020`…, recorded in full in the archive.

### 2. Rounding

All floating-point operations of the verifier run in the rounding mode towards $-\infty$, on nonnegative quantities, so that every computed sum and product is a lower bound for the exact one. The target $\frac{13}{3125}$ is rounded up, $\mu_0/4000$ is rounded down, and the weights $2/(8-r)$ are rounded down.

### 3. Reduction to a compact region

Let $C$ be the least integer with $(\mu_0/4000)C \ge \frac{13}{3125}$; here $C = 53{,}249$. If the gaps lie in cells whose lower ends add up to at least $C$ cells, the linear term alone settles the inequality. On the rest, every single coordinate satisfies $\mu_0 g_i + w(g_i) < \frac{13}{3125}$, since all other terms are nonnegative. Testing this cell by cell leaves two intervals of cells, $[3756, 4877]$ and $[7080, 53248]$, that is gaps in $[0.939, 1.220]$ or in $[1.770, 13.312]$. The $2^7 = 128$ products of these intervals are the initial boxes.

### 4. Lower bound on a box

A box is a product of intervals of cells $[l_i, u_i]$, $i = 1, \ldots, 7$. On it, $\mu_0 \sum g_i \ge (\mu_0/4000)\sum l_i$, and the distance $y_{i+r} - y_i$ lies in the union of the cells from $\sum l$ to $\sum(u+1) - 1$ over the $r$ gaps involved. The minimum of $\underline{w}_c$ over that range of cells, found by a sparse table of range minima, bounds $w$ from below. Adding these bounds with their weights gives a lower bound for $F_8$ on the box. A box is discarded when this bound reaches $\frac{13}{3125}$; otherwise its widest coordinate interval is halved. The verification fails if a box consisting of single cells is not discarded.

### 5. The run

The boxes are first split breadth-first until 222,858 remain, and these are then processed depth-first, in parallel, with the rounding mode set in each thread. The search examined 10,039,560,518 boxes in about a quarter of an hour on two processor cores and discarded all of them. The run for $\mu_0 = \frac{1}{2000}$ and $c_0 = \frac{605}{100000}$, made with the serial version, examined 5,440,924,378 boxes. The same verifier reproduces the inequality of [1] for $R = 7$, $\mu_0 = \frac{1}{3000}$, $c_0 = \frac{19}{5000}$, with the same three intervals of cells.


[1] ainta, A 67.30085% lower bound for simple zeros of the Riemann zeta function, GitHub repository `ainta/zeta-simple-zeros` (2026), mirrored at https://github.com/creelie/zeta-simple-zeros.

[2] L. Alpöge and R. Furman, More than two thirds of the zeta zeros are simple and on the critical line, preprint (2026),

arXiv:2608.13637.

[3] S. A. C. Baluyot, D. A. Goldston, A. I. Suriajaya and C. L. Turnage-Butterbaugh, An unconditional Montgomery theorem for pair correlation of zeros of the Riemann zeta-function, Acta Arith. **214** (2024), 357–376, doi:10.4064/aa230612-20-3.

[4] R. Bhatia, *Matrix Analysis*, Grad. Texts in Math. 169, Springer, New York, 1997, doi:10.1007/978-1-4612-0653-8.

[5] D. Bhattacharjee and U. Bhattacharya, Maximal large sieve and coefficient uniform failure for Liouville correlations, Preprints (2026), doi:10.20944/preprints202606.1350.v1.

[6] D. Bhattacharjee and U. Bhattacharya, BLLM: Beyond the LLM with Deep Human-Mind-AI, preprint (2026), distributed with [8].

[7] D. Bhattacharjee, P. Nandi, O. Frederick and P. Samal, Spectral and analytic structure of the Nyman–Beurling–Báez-Duarte approximation, Preprints (2026), doi:10.20944/preprints202506.0772.v2.

[8] D. Bhattacharjee, Anaxia, the BLLM engine, eBLLM release 3.0.5, Zenodo (2026), doi:10.5281/zenodo.23469389; all versions doi:10.5281/zenodo.23286641.

[9] H. M. Bui, J. B. Conrey and M. P. Young, More than 41% of the zeros of the zeta function are on the critical line, Acta Arith. **150** (2011), 35–64, doi:10.4064/aa150-1-3.

[10] E. Carneiro, V. Chandee, F. Littmann and M. B. Milinovich, Hilbert spaces and the pair correlation of zeros of the Riemann zeta-function, J. Reine Angew. Math. **725** (2017), 143–182, doi:10.1515/crelle-2014-0078.

[11] J. B. Conrey, More than two fifths of the zeros of the Riemann zeta function are on the critical line, J. Reine Angew. Math. **399** (1989), 1–26, doi:10.1515/crll.1989.399.1.

[12] D. R. Heath-Brown, Simple zeros of the Riemann zeta-function on the critical line, Bull. London Math. Soc. **11** (1979), 17–18, doi:10.1112/blms/11.1.17.

[13] F. Johansson, Arb: efficient arbitrary-precision midpoint-radius interval arithmetic, IEEE Trans. Comput. **66** (2017), 1281–1292, doi:10.1109/TC.2017.2690633.

[14] Y. Lamzouri, A new proof that more than $2/3$ of the zeros of the Riemann zeta function are simple and on the critical line, preprint (2026), arXiv:2609.02882.

[15] N. Levinson, More than one third of zeros of Riemann's zeta-function are on $\sigma = \frac{1}{2}$, Adv. Math. **13** (1974), 383–436, doi:10.1016/0001-8708(74)90074-7.

[16] H. L. Montgomery, The pair correlation of zeros of the zeta function, in *Analytic Number Theory*, Proc. Sympos. Pure Math. 24, Amer. Math. Soc., Providence, 1973, 181–193, doi:10.1090/pspum/024/9944.

[17] H. L. Montgomery, Distribution of the zeros of the Riemann zeta function, in *Proceedings of the International Congress of Mathematicians (Vancouver, 1974)*, Vol. 1, Canad. Math. Congress, Montreal, 1975, 379–381.

[18] K. Pratt, N. Robles, A. Zaharescu and D. Zeindler, More than five-twelfths of the zeros of $\zeta$ are on the critical line, Res. Math. Sci. **7** (2020), Paper No. 2, doi:10.1007/s40687-019-0199-8.

[19] A. Selberg, On the zeros of Riemann's zeta-function, Skr. Norske Vid.-Akad. Oslo I **1942**, no. 10, 1–59.

[20] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Clarendon Press, Oxford, 1986.

[21] J. D. Vaaler, Some extremal functions in Fourier analysis, Bull. Amer. Math. Soc. (N.S.) **12** (1985), 183–216, doi:10.1090/S0273-0979-1985-15349-2.

[22] A. Weil, Sur les "formules explicites" de la théorie des nombres premiers, Comm. Sém. Math. Univ. Lund, tome supplémentaire (1952), 252–265.

[23] D. Bhattacharjee and U. Bhattacharya, Simple zeros on the critical line: source, verifier and certificates, GitHub repository `creelie/mh-phys` (2026), `https://github.com/creelie/mh-phys`.